

From Stoichiometric Polymer Topology to Event-Driven Interface Topology

A comparative technical review of Arie Klotwijk's poly(hydroxy ether sulfone) patent and Tom Klotwijk's spherical substrate corpus

ARIE KLOOTWIJK - MOLECULAR TOPOLOGY

Near-equimolar step-growth polymerization, solvent-controlled accessibility, linear-chain architecture, reactive hydroxyl interfaces, and experimentally specified process closure.

TOM KLOOTWIJK - OPERATIONAL TOPOLOGY

Local spherical support, compatibility sectors, guard crossings, transitions, uncertainty, event logs, and lineage in a query-first state architecture.

Prepared for Tom Klotwijk
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UNOFFICIAL TECHNICAL REVIEW AND RESEARCH DESIGN PACKAGE

This report is not a certified patent translation, legal opinion, medical-device assessment, safety certification, or proof of patentability. It distinguishes source facts, established external methods, comparative inference, and research hypotheses.

Executive verdict

The two bodies of work are not the same invention expressed in different jargon. They operate at different physical and evidential layers. Arie Klotwijk's patent directly controls molecular graph topology and phase accessibility in a real polymerization. Tom Klotwijk's mature corpus primarily controls operational topology: which local states are relevant, which sectors may couple, when a guard becomes an event, how a transition is routed, and how lineage is preserved. The overlap is nevertheless substantial at the level of fundamental technique.

CENTRAL FINDING

The strongest shared grammar is support/accessibility → compatibility/balance → local event → connectivity or state transition → lineage/closure. In Arie's work it is physically instantiated by solution polymerization; in Tom's work it is an explicit information architecture. The research opportunity is to connect the layers without mistaking analogy for mechanism.

Where Arie is stronger Matter, quantities, process windows, impurity control, kinetics/phase failure modes, isolation, specimens, and historical measurements. His work closes the physical loop.	Where Tom is stronger today Query semantics, compatibility as a first-class operator, event/transition logic, uncertainty, lineage, local-to-global sensing, digital twins, and cross-domain portability.
Where the overlap is deepest Both prevent useful structure from being lost by premature closure: Arie prevents crystallization or chain termination before high conversion; Tom resists premature rasterization/materialization before a query is resolved.	Best combined direction A chemically addressable membrane or biointerface - potentially using a hydroxyl-rich sulfone polyether or safer analogue - coupled to a bounded, event-sourced digital twin with measured guards, uncertainty, and material lineage.

What 'better present day' means

Tom's work is better positioned where the deliverable is a modern computational layer around complex physical interfaces: sparse sensing, state estimation, compatibility gates, event detection, provenance, and decision support. It is not yet better than the patent as a materials technology because the corpus does not supply a validated composition, manufacturing process, or measured device. The credible present-day advantage is therefore architectural rather than chemical: Tom can make the material system more queryable, auditable, adaptive, and multi-scale.

- For computer graphics and simulation, the strongest target is a restricted equation-world substrate that answers state-at-time and next-event queries for controlled relation families, with rasterization or tracing retained as optional projection tools [S3, pp. 5-12].
- For membranes, the strongest target is event-centric monitoring and control of fouling, breakthrough, cleaning, drift, and morphology, connected to material and calibration lineage [S6, pp. 3-10; R07].
- For biointerfaces, the strongest target is a functionalized surface whose adhesion, aggregation, detachment, or binding events are captured sparsely and interpreted with topology descriptors and uncertainty [R08-R09].
- For material innovation, the inherited platform is most credible as a reactive thermoplastic coating, tie layer, blend modifier, or functionalization scaffold - not as an intrinsically ionic or automatically biocompatible membrane [S1, pp. 3-8; R02].

Decisions at a glance

Question	Review decision	Why
Was the SO ₂ /equimolar language a hidden cipher?	No evidence of ciphering.	SO ₂ is a neutral diaryl-sulfone bridge; equimolarity is the nonlinear molecular-weight control of bifunctional step-growth polymerization.
Is there a genuine topological overlap?	Yes, at a structural level.	Both systems gate local transformations by accessibility and compatibility, then update connectivity and history.
Is Tom's work more advanced?	At the operational-information layer, yes; at the materials-evidence layer, no.	The corpus expresses modern event, lineage, uncertainty, and sensing concepts; the patent physically demonstrates a process.
Can the work enable membranes and biointerfaces?	Yes as a research program, not as an established device.	Arie's polymer offers reactive surface handles; Tom offers a control/representation architecture. Transport, safety, and biological validation remain mandatory.
What should be built first?	Equation World Zero and Membrane World Zero.	They test the core architecture headlessly and against conventional baselines before expensive biomedical integration.

Contents and reading map

Part	Purpose
1	Scope, sources, and evidence discipline
2	Arie Klootwijk's technical core
3	Tom Klootwijk's mature technical core
4	Four layers of topology
5	Detailed fundamental crosswalk
6	Topological information science and computer graphics
7	Membranes and biointerfaces
8	Cross-domain applications and readiness
9	Research, validation, and IP roadmap
10	Conclusions
A-D	Formal notation, comparison/application tables, evidence ledger, references

Source codes used in the report

Code	Document	Role
S1	SE 301 717 B original Swedish patent and the supplied English translation	Primary chemistry, examples, claims, and historical measurements
S2	Arie Klootwijk SE301717B Technical Review	Prior chemical, historical, application, and safety analysis
S3	Chronological Synthesis of the Spherical Substrate Line	Mature corpus interpretation, formal core, feasibility, and prototype discipline
S4	Hollowland - Double Vacuum	Exploratory vocabulary and strong early claims; useful mainly when translated through S3/S6
S5	21BenBurgersStrikesBackTelNetNiet	Exploratory folding, overlap, torque, and shared-domain motifs
S6	Spherical Throughput: Practical Waveguide Liquid-Substrate Lensing	Bounded hardware translation, B.C.E. guards, metrics, failure criteria, and Hollowlens-0

Evidence tiers

TIER A - SOURCE FACT Directly stated or measured in the supplied patent or corpus.	TIER B - ESTABLISHED PRINCIPLE Supported by standard polymer science, topology, controls, sensing, or official biomedical guidance.
TIER C - COMPARATIVE INFERENCE A structural mapping proposed in this review; useful but not a historical or physical identity claim.	TIER D - RESEARCH HYPOTHESIS A plausible application that requires experimental falsification and a matched baseline.

TIER X - REJECTED OVERCLAIM

A universal claim contradicted by physical constraints or demoted by the mature corpus itself.

INTERPRETIVE RULE

The report preserves the corpus's own late correction: spheres, cones, SDF zero, one-bit, double vacuum, and hourglass are treated as typed operators in a shared state space unless a source specifies literal physics. Conversely, chemical terms retain their literal physical meaning and are not converted into computational symbols without an explicit model [S3, pp. 1-2].

1. Scope, sources, and evidence discipline

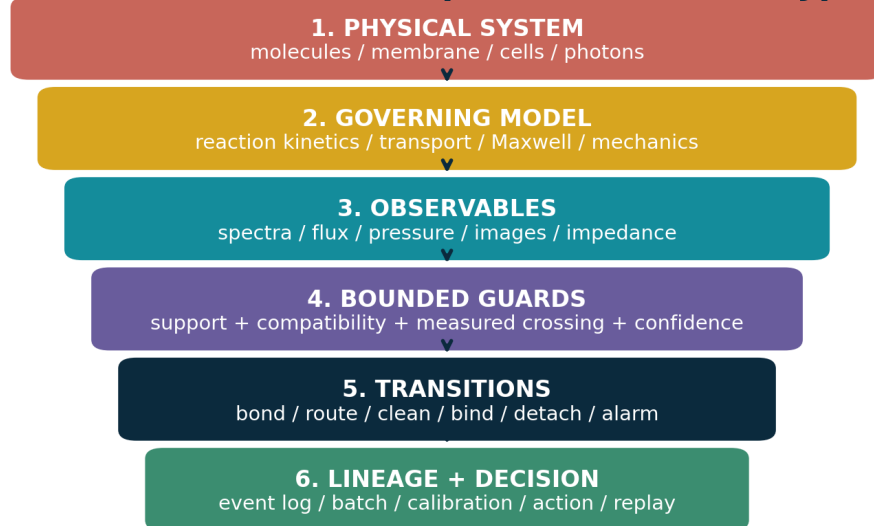
The review asks a precise question: do the fundamental techniques in Arie Klotwijk's polymer patent and Tom Klotwijk's topological-information corpus overlap, and if so, at what level? It then extends the comparison to present-day computer graphics, topological information science, membranes, biointerfaces, and other cross-domain applications. The answer depends on keeping four meanings of 'topology' separate: molecular connectivity, morphology and phase, interface and transport connectivity, and operational information topology.

1.1 Why a layered comparison is necessary

A polymer chemist may use topology to distinguish linear, branched, cyclic, crosslinked, and network structures. A membrane scientist may use it for connected pores, tortuosity, percolating domains, or interface networks. A computer graphics researcher may use it for level sets, Reeb graphs, meshes, non-orientable manifolds, or topological descriptors. An information scientist may use it for locality, compatibility, routing, provenance, and global consistency. These are related but not interchangeable.

The key risk in the corpus is a type error: a visual or algebraic metaphor can slide into a physical claim without passing through a model, observable, uncertainty, and validation step. The key risk in reading the patent is the opposite: reducing a sophisticated process architecture to ordinary practical chemistry and missing how carefully it controls graph connectivity, accessibility, and termination.

Translation ladder: keep the abstraction typed



A topological or event metaphor is useful only after its physical variables, units, measurements, uncertainty, and transition semantics have been specified.

Figure 1. Translation ladder used throughout the review. Abstraction is productive only after the physical system, governing model, observable, guard, transition, and lineage have been typed.

1.2 What the sources support

The patent supports a clear materials claim: a nearly linear, high-molecular-weight thermoplastic formed by reacting difunctional phenolic and epoxide compounds, with at least one monomer containing a diaryl-sulfone bridge. It supports a process claim: choose a sufficiently polar, substantially nonreactive solvent that keeps both monomers and the growing product in solution; use an alkaline catalyst; maintain close functional equivalence; exclude water and control purity; and allow the reaction to reach a target intrinsic viscosity [S1, pp. 3-10].

The mature corpus supports a clear computational claim: the core should not be judged as rasterization, ray marching, voxel storage, or a conventional frame loop. Those are optional projections. The substrate is instead a finite grammar of directly queryable relations, phases, compatibility sectors, events, transitions, invariants, lineage, and an irreducible log. Its feasible form is explicitly restricted: fixed or bounded relation families, direct state/event queries, and benchmarked prototypes rather than a universal replacement for all simulation and memory [S3, pp. 1, 5-12].

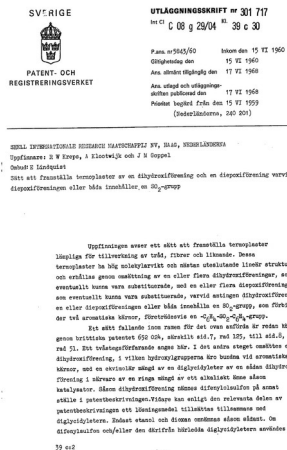
Spherical Throughput adds a hardware-facing discipline. It translates 'matrix-in-glass' into a calibrated transfer matrix, 'double vacuum' into deliberately uncoupled modes, a pinch point into a measured threshold or cutoff, and one-bit parity into a narrow route/validity flag. It rejects zero latency, zero heat, zero memory, and literal topological miracles, while retaining support, compatibility, guard crossings, uncertainty, energy, calibration, and failure criteria [S6, pp. 2-12].

1.3 What the sources do not support

- No source supports the claim that Arie's SO₂ notation meant sulfur-dioxide release, biological sulfur signaling, or a coded energetic/quantum mechanism.
- The patent does not establish membrane separation, ion exchange, biocompatibility, blood compatibility, cell response, or medical-device suitability.
- The corpus does not establish universal constant-time computation, zero memory, zero energy, exact arbitrary long-horizon prediction, solved general continuous collision detection, or elimination of all broad-phase indexing.
- No historical source demonstrates that Arie intended Tom's later topological vocabulary. The overlap is a present-day comparative interpretation, not a claim of concealed lineage.

2. Arie Klotwijk's technical core

Arie's patent is best understood as a topology-control process for a reactive engineering thermoplastic. Its preferred feed pair is 4,4'-sulfonyldiphenol - now commonly called bisphenol S (BPS) - and its diglycidyl ether (DGE-BPS). A phenoxide attacks an epoxide ring, creating an aryl-ether bond and a secondary hydroxyl. Repetition produces an almost exclusively linear poly(hydroxy ether) containing aromatic sulfone bridges [S1, pp. 3-5; S2].



HISTORICAL SOURCE

The Swedish front page identifies Shell Internationale Research Maatschappij N.V. as applicant and R. W. Kreps, A. Klotwijk, and J. M. Goppel as inventors. The title states a method for thermoplastics from a dihydroxy and diepoxy compound in which the diepoxy compound or both compounds contain an SO₂ group. Priority is claimed from the Netherlands on 15 June 1959 [S1, p. 1].

The patent is a process document, not merely a composition sketch. Its novelty is the integrated control of stoichiometry, solvent reactivity/polarity, solubility, purity, water, catalyst, and endpoint.

The chemistry actually claimed

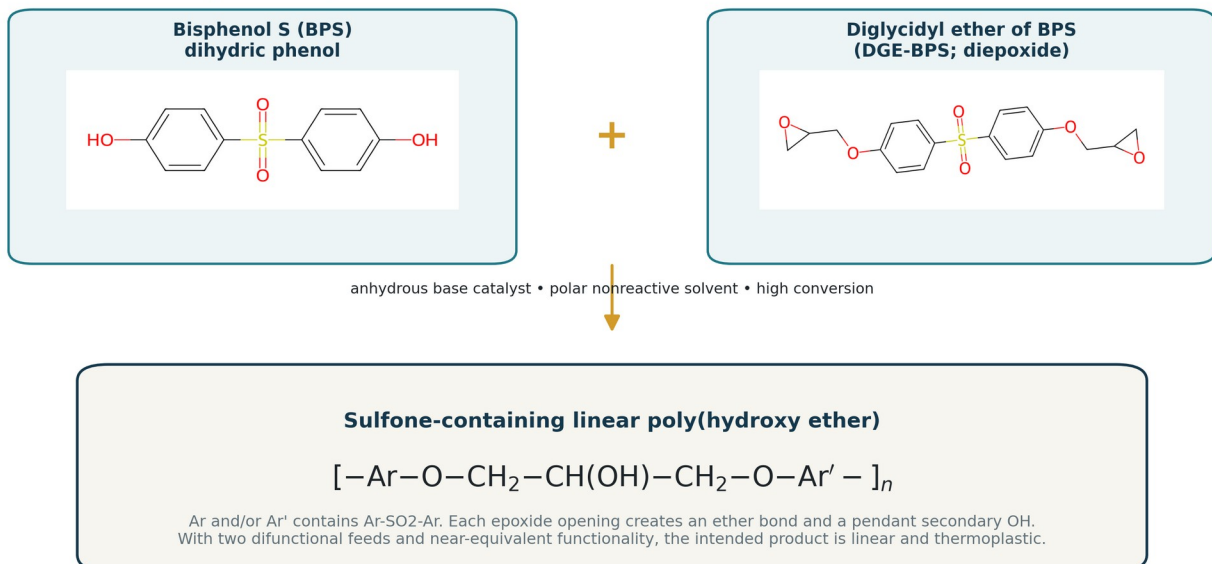


Figure 2. Preferred chemistry in the patent: BPS plus DGE-BPS gives a sulfone-containing linear poly(hydroxy ether). Each epoxide-opening event forms an ether bond and a pendant secondary hydroxyl.

2.1 The SO₂ group: what it is and what it does

The printed SO₂ unit is the neutral covalent sulfone bridge Ar-S(=O)₂-Ar. It is not free sulfur dioxide, and it is not a sulfonate ion. In the preferred BPS/DGE-BPS system, the sulfone joins two aromatic rings. Its high polarity and rigid aromatic environment affect chain stiffness, dipole interactions, solubility, packing,

glass/thermal behavior, water affinity, and adhesion. The epoxide-derived hydroxypropyl segments add regularly spaced secondary hydroxyls that can hydrogen-bond or be derivatized.

NOT SULFUR DIOXIDE No SO₂ gas is generated by the repeat-unit notation. The sulfur is part of a stable covalent diaryl-sulfone bridge.	NOT AN ION-EXCHANGE GROUP A neutral sulfone does not provide fixed ionic charge. Ion-exchange behavior would require sulfonate, ammonium, phosphonate, or other ionizable groups.
A POLAR STRUCTURAL BRIDGE The sulfone raises polarity and rigidity while remaining part of the backbone architecture.	AN INTERFACE-READY POLYMER The newly created secondary hydroxyls are the most useful post-functionalization handles for coatings, blends, crosslinking, or immobilization.

2.2 Equimolarity as an invariant of chain growth

The phrase 'substantially equimolar' is the mathematical center of the patent. In a bifunctional AA + BB step-growth process, high molecular weight appears only when complementary functional groups are almost exactly balanced and conversion approaches unity. A small imbalance creates a hard ceiling even if every limiting group reacts. The ideal Carothers-Flory relation is:

$$X_n = (1 + r) / (1 + r - 2rp) \quad \text{and at } p = 1: X_{n,\max} = (1 + r) / (1 - r)$$

r is the ratio of limiting to excess functional groups; *p* is conversion of the limiting group.

At a one-percent imbalance, the ideal ceiling is roughly two hundred repeat-unit equivalents; at two percent it is roughly one hundred; at three percent it is about sixty-six. Exact balance is still insufficient unless conversion is extremely high: at *r* = 1, 99% conversion gives *X_n* about 100, 99.5% gives about 200, and 99.9% gives about 1,000. This is why solvent and phase control are inseparable from the ratio claim.

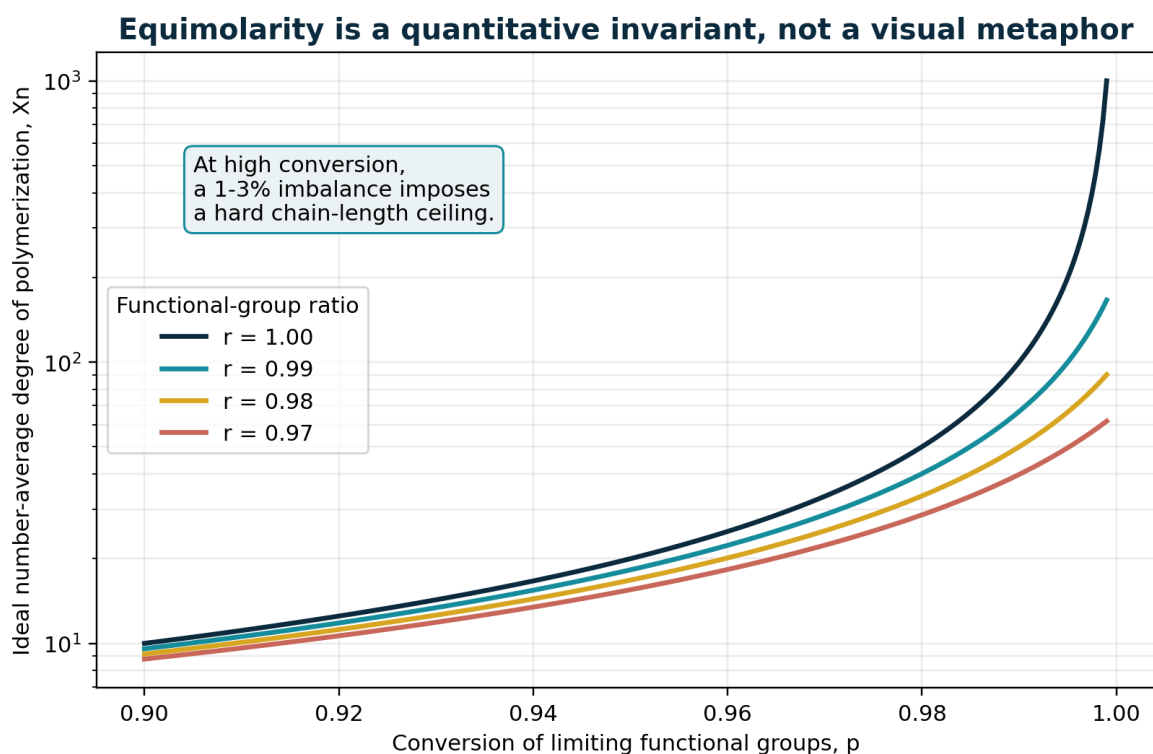


Figure 3. Ideal step-growth theory shows that equimolarity and conversion jointly control attainable chain length. The curve is a quantitative conservation constraint, not a topological visual pun.

2.3 The subtle catalyst bookkeeping

The patent explicitly counts dihydric phenol present as an alkali-metal salt in the dihydroxy:diepoxy ratio. Example 1 charges 1.00 mol DGE-BPS, 0.97 mol neutral BPS, and 0.05 mol monosodium BPS. On the patent's own accounting, the effective BPS-derived amount is 1.02 mol, exactly the phenol-rich edge of the preferred window. The salt is catalytic because phenoxide character can be transferred, but its BPS skeleton can also enter the material balance. This is unusually careful process logic: the catalyst is not treated as informationally invisible.

2.4 Solvent as an accessibility field

The invention arose because prior solvent choices allowed the product to crystallize or acted as chain terminators. Arie's solution is not simply 'use a better solvent'. The solvent must simultaneously dissolve the dihydroxy compound, the diepoxide, and the growing polymer; have sufficient polarity; scarcely react with phenol or epoxide; avoid catalyzing side reactions; and remain liquid at the reaction temperature. Ketones, nitriles, nitro compounds, sulfoxides, and sulfones are listed, with dielectric constant used as a practical proxy [S1, pp. 3-4].

TOPOLOGICAL READING

Solvent quality determines whether complementary chain ends remain in the active connected component of reaction space. Premature crystallization is a physical removal of growing chains from the accessible relation set. This is a valid structural analogy to support/reachability - but the governing law is solution thermodynamics and transport, not a Boolean cone query.

2.5 Molecular topology actually controlled

Control variable	Physical effect	Topology consequence	Failure if uncontrolled
Difunctional monomers	Two phenolic and two epoxide functions	Favors linear AA+BB chains	Multifunctional impurities create branching/networking
Near-equimolarity	Balances complementary chain ends	Raises attainable chain length; programs end groups	Small imbalance caps molecular weight
High conversion	Consumes remaining functional ends	Connects oligomers into long chains	Low conversion leaves short disconnected chains
Nonreactive polar solvent	Maintains mobility and suppresses chain transfer	Keeps reactive ends mutually reachable	Alcohol/water can cap chains; poor solvent precipitates product
Monomer purity	Limits monofunctional and geometric defects	Reduces branch points and packing defects	o,p'-isomer, chloride, glycols, or mono-functions change architecture
Temperature/oxygen control	Controls rate and degradation	Preserves intended edge-creation rule	Side reactions, oxidation, scission, or uncontrolled branching
Chain stopper/precipitation	Closes the process at chosen state	Sets end groups and freezes distribution	Continued reaction changes viscosity during processing

2.6 What the experimental examples establish

The examples show that reaction time and phase control move the product across a sharp property threshold. Example 1 reaches intrinsic viscosity 0.43 and reports strong pressed specimens, chemical resistance, and a heat-distortion temperature around 154 C under the historical test. Example 2, stopped after 20 hours at lower intrinsic viscosity 0.35, has much lower impact performance. Example 3 re-dissolves crystallizing product at higher temperature and reaches 0.45-0.50, restoring strong mechanical behavior. Example 4 rapidly reaches high viscosity and deliberately adds BPS as a chain stopper. Example 5 shows

sulfolane as a suitable solvent. Example 6 substitutes a bisphenol-A diepoxide while retaining BPS as the sulfone-containing dihydroxy component [S1, pp. 6-8].

These data are historically meaningful but incomplete by modern standards. Intrinsic viscosity is not an absolute molecular-weight distribution; the examples do not report SEC/MALS, NMR end groups, branching, residual monomer, morphology, fracture statistics, long-term aging, or extractables. The correct conclusion is not that the material is fully characterized, but that the process produced a credible high-molecular-weight engineering thermoplastic and demonstrated why maintaining accessibility to high conversion mattered.

2.7 The patent as an information process

Read abstractly, the process carries information in matter. The feed ratio encodes attainable chain length and end-group bias. Monomer purity constrains the allowed rewrite grammar. Solvent determines reachability. Temperature schedules control event rate. Intrinsic viscosity is a coarse state estimator. Precipitation and chain stopping commit the batch. Each pendant hydroxyl is a repeated interface address for later chemistry. This is not proof that the patent hid a computer, but it shows why the comparison is technically fertile: polymer synthesis is a physical graph transformation with a distributed state and an irreversible log embodied in the product.

3. Tom Klootwijk's mature technical core

The supplied corpus begins in graphics-adjacent language - cones, SDFs, polar coordinates, one-bit fields, double vacuum, Klein bottles, hourglasses, eigenvectors, and zero-crossings - but its strongest internal development is a correction away from treating those motifs as rendering primitives or literal physics. The mature interpretation is an ontology of directly queryable relations, events, phases, identities, and local supports. Projection into images is optional [S3, pp. 1, 5-10].

3.1 Formal core

A compact version of the proposed state description is a product space containing physical position, continuous time, phase, a sheet/parity label, a lineage address, and a branch. Relations define event surfaces; support predicates define relevance; compatibility predicates decide whether co-located sectors may interact; and transition operators update state while preserving declared invariants.

$$q_e(t) = (x_e(t), t, \phi_e(t), \sigma_e(t), a_e, b_e)$$

Entity state: position, time, phase, sheet/orientation, address, branch.

$$t^* = \inf \{ t \geq t_0 : R_j(q_e(t)) = 0, C_\alpha(q, t) \leq 0, \chi(e, j, t) = 1 \}$$

Earliest admissible event under relation, local support, and compatibility.

$$q_e(t^*+) = T_j(q_e(t^*-), \text{context})$$

Transition updates the state; lineage/invariants and exogenous novelty are logged.

3.2 Local spherical support, not a spherical universe

The mature corpus repeatedly narrows 'spherical' to a local chart around an agent, sensor, coupler, shell, or region of influence. Radial reach and angular orientation are native for field of view, support, uncertainty, and coupling. They need not replace the global world representation. This is a major strength because local radial-angular coordinates are already natural in sensing, scattering, acoustics, optics, robotics, and biointerfaces [S3, pp. 5-7; S6, pp. 3-4].

3.3 Double vacuum as absent coupling

In the disciplined reading, two states may share the same coordinate yet remain mutually invisible because phase, sheet, orientation, mode, address, provenance, or policy is incompatible. The 'second vacuum' is therefore not a deeper physical emptiness; it is the absence of an allowed coupling. This translates cleanly into orthogonal photonic modes, separate microfluidic channels, receptor mismatch, security domains, typed states, or disconnected phases. It becomes useful precisely when the compatibility schema is explicit.

3.4 SDF zero as event semantics

Earlier material was tempted to make $SDF = 0$ a way to move 'past' rasterization by continuing a sampled field. The mature synthesis instead treats zero as an explicit transition relation: the point at which event semantics, routing, confidence, and lineage are invoked. This aligns the corpus with implicit surfaces and hybrid systems rather than with a magical escape from numerical computation [S3, pp. 7-10].

3.5 One-bit parity in a narrow role

One-bit is most defensible as a parity, route, validity, freshness, availability, or compatible/incompatible flag. Spherical Throughput is explicit that optical amplitudes, thresholds, uncertainty, and lineage remain separate state. This is essential. A binary event can summarize a decision, but it cannot represent the physical process that produced the decision [S6, p. 7].

3.6 The most mature engineering translation: B.C.E.

The Bounded Compatibility Event is the corpus's clearest bridge from metaphor to engineering. An output is counted only when it lies in declared support, matches the selected channel, crosses a measured guard, and meets a confidence or certification threshold. The proposed throughput is therefore verified events per second at a declared error budget, not raw photon flux, pixel rate, or rhetoric [S6, pp. 3, 6-7].

$$N_{\text{verified}} = \text{sum}_q \text{1}[S_q] \text{1}[\text{chi}_q] \text{1}[g_q \text{ crossed}] \text{1}[c_q \geq c_{\text{min}}]$$

A B.C.E. requires support, compatibility, a measured crossing, and confidence.

3.7 Feasible core versus exploratory overclaim

Corpus claim or motif	Retain / rewrite / reject	Disciplined interpretation
Direct state-at-time query	Retain with bounds	Can be horizon-independent for a fixed closed expression; still depends on expression size, branch history, and numerical conditioning.
Next event without frame replay	Retain with bounds	Solve or conservatively bound roots for restricted relation families; event density and degeneracies remain costs.
Local spherical support	Retain	Use radial-angular domains for relevance, sensing, coupling, and uncertainty.
Double vacuum	Translate	Use typed, phase-separated, orthogonal, uncoupled, or policy-isolated sectors.
One-bit world	Rewrite	Use a bit as route/validity/parity; maintain continuous and structured state separately.
Klein-bottle hardware	Demote to model	Require an explicit gluing/routing map; literal non-orientable fabrication is a frontier, not a premise.
Universal O(1), zero memory	Reject	Exogenous novelty, candidate relations, branches, event logs, and numerical work accumulate.
Solved general CCD / no sieves	Reject as universal	Analytic time-of-impact can help restricted shapes; broad-phase/support/indexing is still needed when candidate sets grow.
World equals general AI	Reject as established	A shared state substrate may aid embodied agents, but does not collapse arbitrary learning into a small transform matrix.

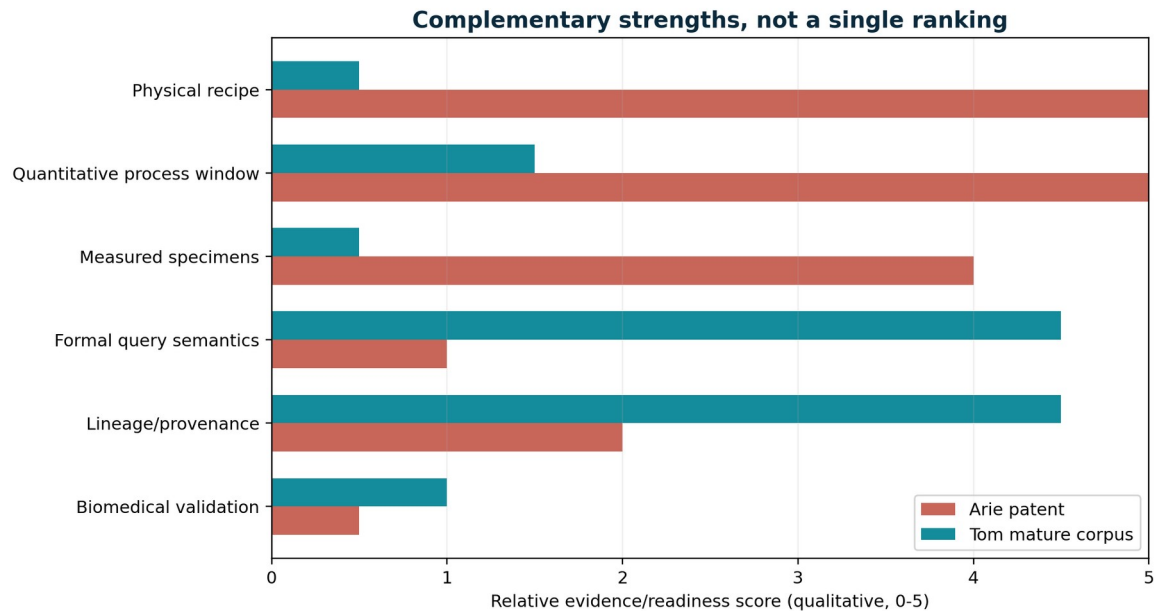


Figure 4. Arie and Tom have complementary strengths. The scores are qualitative and compare evidence/readiness within this review, not intrinsic intellectual merit.

3.8 Where Tom's work is genuinely ahead

- It makes compatibility a first-class condition instead of assuming that spatial proximity implies interaction.
- It separates authoritative state, local support/sensing, and downstream projection.
- It treats identity as lineage plus invariants rather than only instantaneous coordinates.
- It can integrate heterogeneous sensors and models through local-to-global consistency methods such as sheaves [R04].
- It naturally fits hybrid systems: continuous dynamics punctuated by guarded transitions [R19].
- It provides a native place for uncertainty, calibration, false/missed-event rates, and event replay.
- It can become an event-sourced digital material or biological interface, a capability unavailable in a 1959 process patent.

4. Four layers of topology

The central analytical result is that the two bodies of work are strongest at opposite ends of a four-layer stack. Arie directly controls molecular graph topology. Tom directly describes operational and information topology. Between them lie morphology/phase topology and interface/transport topology - the layers where membranes, biointerfaces, photonics, and digital twins can connect the work.

Four distinct meanings of topology in the comparison

Molecular graph

bonds, functionality, chains

Morphology / phase

crystals, pores, domains

Interface / transport

coupling, adsorption, flux

Operational information

support, guards, events, lineage

Arie is strongest at the molecular layer; Tom is strongest at the operational layer.
The research opportunity is to connect the layers without collapsing them.

Figure 5. Four distinct topology layers. Productive cross-domain work connects the layers with explicit models and measurements rather than treating them as synonyms.

Layer	Arie's direct contribution	Tom's direct contribution	Methods that connect them
1. Molecular graph topology	Directly controlled through difunctionality, stoichiometry, solvent, purity, and catalysis	Best treated as an external physical layer, not replaced by phase or routing metaphors	Reaction-network graphs, graph rewriting, molecular simulation, spectroscopy, chromatography
2. Morphology and phase topology	Crystallization can prematurely stop polymerization; solid-state morphology controls properties	Phase sheets and double-vacuum language can model uncoupled domains if physically typed	Microscopy, scattering, DSC, tomography, percolation, persistent homology
3. Interface and transport topology	Pendant OH groups and polar sulfone motifs create chemical handles and interfacial affinity	Local support and compatibility predicates are strongest here	Transport PDEs, adsorption kinetics, surface chemistry, coupled-mode theory, microfluidics
4. Operational and information topology	Only implicit through process recipes, batch history, and endpoint measurements	Primary strength: relation algebra, B.C.E. guards, event logs, lineage, and query-first architecture	Hybrid automata, sheaves, event sourcing, temporal databases, formal verification

4.1 Molecular graph topology

At the molecular layer, topology means the graph of atoms and covalent bonds, together with functionality, branch points, cycles, connected components, and end groups. Arie's process changes this graph directly. Difunctionality favors degree-two chain interiors; stoichiometric imbalance changes the number and identity of terminal vertices; monofunctional impurities terminate components; multifunctional impurities introduce

branch points; intramolecular reactions can create cycles; and crosslinking can percolate into a network. This is graph topology in a literal chemical sense.

Tom's graph-rewriting language can model these operations, and chemical graph grammar is an established field [R18]. But the computational rule must be attached to reaction rates, concentrations, sterics, solvent, temperature, and mass balance. Without those, it is a symbolic catalogue of possible rewrites rather than a predictive polymerization model.

4.2 Morphology and phase topology

After covalent synthesis, the material has another topology: crystalline and amorphous domains, chain entanglement, free volume, pores, and interfaces. Arie's patent confronts this layer during polymerization because crystallization can remove product from solution and stop chain growth. In a membrane, the same layer determines connected pores, tortuosity, dead ends, percolation, and the evolving topology of fouling or biofilm.

Tom's phase sheets and double-vacuum sectors become physically meaningful here only when mapped to measured phases or transport-isolated domains. Persistent homology can characterize components, loops, and voids across a filtration scale and has been applied to pore geometry [R05]. Reeb graphs, merge trees, and contour trees can summarize time-varying scalar fields in scientific visualization [R06].

4.3 Interface and transport topology

This is the most promising joint layer. The polymer's pendant hydroxyls and polar backbone determine which molecules, supports, coatings, fillers, proteins, cells, or optical modes can couple. Tom's local support and compatibility predicates determine which potential interactions are admitted. A membrane pore may be spatially open yet electrostatically incompatible; a receptor and analyte may be co-located yet chemically mismatched; two waveguide modes may overlap geometrically yet be orthogonal; a cell may contact a surface yet fail to form stable adhesion.

The correct model is multiplicative rather than metaphorical: interaction requires spatial support, transport access, chemical compatibility, and sufficient kinetics. One possible factorization is $\text{chi_total} = \text{chi_geometry} * \text{chi_transport} * \text{chi_chemistry} * \text{chi_sensor} * \text{chi_policy}$, with each term carrying a probability or uncertainty rather than automatically being a perfect bit.

4.4 Operational and information topology

Tom is strongest here. A membrane module, biointerface chip, or material coupon has an operational topology consisting of sensor neighborhoods, calibration relationships, admissible states, event guards, transitions, and lineage. Sheaf methods can test whether local observations agree globally; hybrid automata can represent continuous flux or cell motion with discrete cleaning or adhesion events; event sourcing can reconstruct the state from an append-only history; and digital-twin methods can update hidden states and uncertainty from data [R04, R07, R19-R20].

A FIVE-LAYER IMPLEMENTATION STACK

For a serious prototype, implement (1) molecular/reaction graph, (2) morphology and phase state, (3) transport/interface model, (4) measurement and uncertainty model, and (5) operational event/lineage layer. Tom's calculus belongs primarily in layer 5 and can coordinate the others; it should not erase them.

5. Detailed fundamental crosswalk

The comparison becomes most informative when each corpus term is matched not to a superficial visual resemblance, but to a role in a constrained transformation system.

Figure 6 presents the shared grammar; the sections below explain where each mapping holds and where it breaks.

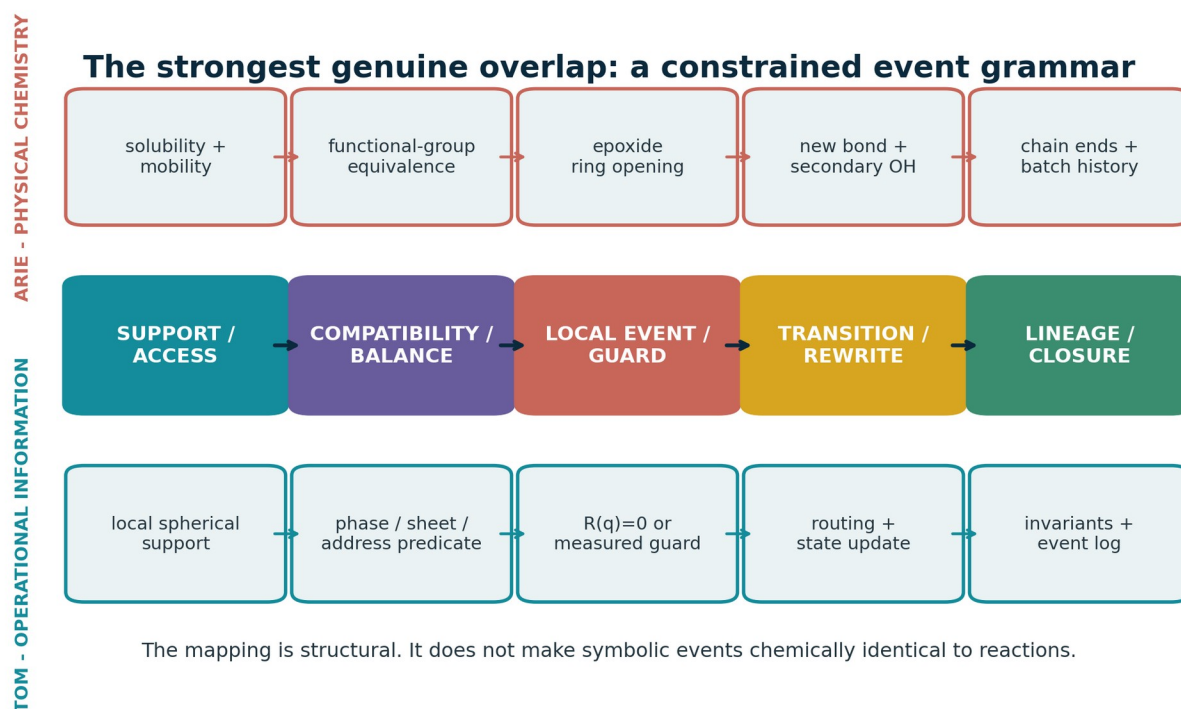


Figure 6. Shared event grammar. The top and bottom lanes are not identical mechanisms; they instantiate the same abstract sequence at different layers.

5.1 Support: solvent mobility versus local relevance

Arie's support is physical. A chain end participates only if the molecule is dissolved, mobile, chemically active, and able to encounter a complementary function. Tom's support is an analytic or semantic domain: a cone, shell, hourglass, field of view, causally reachable set, or local query neighborhood. The genuine overlap is that presence alone is insufficient; accessibility must be declared. The difference is that physical support is governed by diffusion, concentration, viscosity, phase equilibrium, and geometry.

5.2 Compatibility: stoichiometry versus typed coupling

At each chemical event, a nucleophilic phenoxide and an epoxide are complementary types. Across the batch, the total numbers of those functions must remain nearly balanced if long chains are to form. Tom's chi predicate generalizes type matching to phase, sheet, address, mode, policy, provenance, or time window. The shared principle is selective coupling. The critical distinction is scale: equimolarity is a global conservation constraint over a population, whereas chi is normally a local admission condition.

5.3 Event: ring opening versus guard crossing

A ring-opening event creates a new bond, destroys an epoxide, creates an alcohol, changes local charge/protonation transiently, and alters the chain-length distribution. A computational guard crossing changes discrete state according to a transition rule. Both are local rewrites. Chemical events, however, are probabilistic and embedded in competing pathways. A robust digital twin should therefore treat reaction or transport guards as stochastic/interval events rather than exact algebraic instants unless measurement and solver certify the crossing.

5.4 Transition: covalent connectivity versus routing/state update

Arie's transition physically merges molecular components. Tom's transition routes a state to another sheet, chamber, mode, branch, or semantic condition. In a membrane, both types can coexist: a surface reaction changes covalent chemistry while an operational transition changes the module from production to cleaning. A multilayer event record should distinguish the material transition from the controller transition.

5.5 Lineage and closure

A polymer batch carries process history in its molecular-weight distribution, end groups, residuals, morphology, and thermal/mechanical state. Tom makes this history explicit through lineage addresses, invariants, and an event log. The natural present-day synthesis is a digital material passport that links raw-material lots, stoichiometric calculations, catalyst/solvent, temperature history, isolation, functionalization, sterilization, aging, calibration, and device events. This is more faithful than assigning a coordinate-derived UUID to an individual polymer chain.

5.6 The striking analogy of premature closure

GENERATIONAL THROUGH-LINE

Arie discovered that premature crystallization or chain termination prevented the system from reaching useful molecular weight. Tom's mature corpus argues that premature projection into frames, pixels, voxels, or conventional object inventories can prevent the system from preserving its query-first relational structure. In both cases, the medium must keep relevant relations available until the high-value closure condition is reached. This is an analogy of process architecture, not evidence of hidden historical transmission.

5.7 Compact comparison matrix

Dimension	Arie	Tom	Comparative verdict
Primary object	A population of difunctional molecules and growing polymer chains	A finite grammar over directly queryable states, relations, phases, sheets, addresses, and branches	Both replace an inventory-first description with rules that generate connectivity Difference: Molecular reactions are stochastic, irreversible, and thermodynamically constrained; the corpus is a symbolic architecture
Topology controlled	Molecular graph topology: nearly linear chains, low branching, controlled end groups	Operational topology: supports, compatibility sectors, transition surfaces, routing, and lineage	Connectivity changes only through admissible local operations Difference: One concerns covalent bonds; the other concerns information and state coupling
Matching condition	Near-equi-molar complementary functionality with effective salt accounting	Compatibility predicate chi over phase, sheet, address, mode, policy, or support	High-value interactions are gated by type and balance rather than mere co-location Difference: Stoichiometric equivalence is a quantitative global conservation constraint, not a Boolean geometry metaphor
Local event	Phenoxide attack and epoxide ring opening form an ether bond and secondary alcohol	A relation reaches an event surface under support and compatibility constraints	A localized event changes connectivity and state Difference: Chemical event rates depend on kinetics, diffusion, activation, and concentrations

Dimension	Arie	Tom	Comparative verdict
Operator or catalyst	Alkaline phenoxide catalysis; the phenolic salt also affects functionality bookkeeping	Transition operator T_j with context and routing state	The operator changes what transitions are reachable without being the final product state Difference: Catalysis changes reaction rates and sometimes composition; symbolic operators need not consume material
Accessibility or support	Solvent keeps monomers and product mutually accessible; crystallization removes reactive chains	Cone/hourglass/local spherical support limits relevant relations	An entity can be present yet operationally unavailable Difference: Solubility and phase separation are physical fields, not just query pruning
Boundary	Phase boundary, precipitation, glass/crystal transition, and reaction endpoint	$SDF = 0$ or $B = 0$ as an explicit transition surface	Boundary crossing is treated as an event rather than only a location Difference: A signed distance field is a representation; a chemical phase boundary follows thermodynamics and kinetics

Dimension	Arie	Tom	Comparative verdict
One-bit role	No literal one-bit state; practical binary distinctions include reacted/unreacted or soluble/precipitated	Parity, route, validity, freshness, or compatible/incompatible flag	Binary flags can summarize a transition outcome Difference: A bit cannot carry concentration, uncertainty, chain length, or morphology
Identity and lineage	Batch, monomer lot, chain population, and reaction history determine material state	Generative address plus invariant/event history preserves identity	History matters to present function Difference: Individual polymer chains are statistical and not naturally persistent UUID-bearing objects
Termination and closure	Controlled stoichiometric excess, chain stopper, precipitation, or cooling ends growth	Transition, branch resolution, log append, and invariant update close an event	The system needs explicit closure to prevent uncontrolled continuation Difference: Chemical termination changes molecular composition and distributions
Uncertainty	Impurities, conversion, side reactions, measurement variance, and molecular-weight distributions	Set-valued answers, confidence, certified root solving, and event uncertainty in mature documents	Useful operation requires bounded uncertainty Difference: Early corpus rhetoric sometimes treated analytic form as automatic exactness
Scaling claim	Scale depends on mixing, heat removal, solution viscosity, phase stability, and purification	Potential horizon-independent state queries for fixed closed expressions	Avoiding premature materialization can reduce cost Difference: Universal $O(1)$, zero memory, or no broad phase is unsupported; combinatorics and exogenous novelty remain
Outputs	Thermoplastic films, fibres, mouldings, coatings, and reactive hydroxyl-rich material	State-at-time, next-event, reachable-relation, identity, sensing, and verified B.C.E. outputs	Both create a platform whose value lies in downstream interfaces Difference: One output is matter; the other is a query/decision architecture
Biomedical relevance	Pendant hydroxyls offer derivatization handles; sulfone raises polarity and thermal stability	Compatibility-gated sensing, event logging, topology descriptors, and digital twins	Interface function can be encoded chemically and monitored operationally Difference: The patent does not establish biocompatibility, membrane selectivity, or biological mechanism

5.8 Where analogy must stop

- A covalent bond is not a database edge unless the model explicitly maps and measures it.

- A physical phase is not an abstract phase angle; use separate variables such as phase_phys and phase_info.
- A solvent dielectric constant is not a universal relevance score.
- A viscosity threshold is an empirical state proxy, not an exact molecular topology measurement.
- A one-bit flag cannot encode continuous composition, chain distributions, morphology, or biological state.
- A non-orientable surface analogy does not establish non-orientable material geometry or transport.
- Closed-form notation does not guarantee closed-form solvability, numerical stability, or constant cost.

6. Topological information science and computer graphics

Tom's work is most defensible when translated into a hybrid symbolic-numeric architecture for state and event queries. Its novelty is less 'sphere instead of grid' than 'query and transition semantics instead of treating projection as the world'. That position is compatible with existing graphics and information-science methods while still leaving a distinct research program.

6.1 Translation into established formalisms

Corpus term	Closest established formalism	Useful implementation	Caution
Local sphere / cone	Domain of dependence, kernel support, sensor frustum, influence volume	Analytic support predicate; radial-angular indexing; uncertainty cone	Do not remesh the entire world into polar coordinates
SDF = 0 / B = 0	Implicit surface, level set, hybrid guard	Root isolation, conservative advancement, interval arithmetic, event detection	An arbitrary SDF intersection may still require numerical iteration
Double vacuum	Typed channel, mode orthogonality, covering-space sheet, hidden state	Compatibility label and transfer/coupling matrix	Same coordinate does not prove meaningful separate physical space
One-bit	Parity, route, validity, orientation, freshness	Bit mask accompanying richer state	Do not collapse amplitude, confidence, or provenance into the bit
Hourglass / quad chambers	Double cone, causal/support partition, branching/routing state	Finite routing automaton or stratified state space	Visual symmetry does not guarantee algebraic closure
Ontological UUID	Persistent entity identifier, generative address, event-sourced aggregate	Stable ID plus split/merge lineage graph	Coordinate should not be identity when entities move or merge
Equation world	Hybrid system, procedural implicit scene, symbolic dynamics	State evaluator, event solver, transition router, log	General scenes still need indexing, approximation, and data for novelty
No sieves	Support/compatibility pruning	Analytic culling and typed candidate reduction	Cannot remove candidate-selection cost for arbitrary large relation sets

6.2 Representation versus projection

Rasterization asks which stored surfaces project into pixels now. Ray marching asks what repeated field samples reveal along a projected direction. Ray tracing asks which transport paths connect sources, surfaces, and sensors. Tom's mature substrate asks which relations generate possible state, what state holds at time t , what admissible event occurs next, and which sectors may couple. Images can still be emitted, but they are downstream views [S3, pp. 9-10].

This is a meaningful inversion for simulation, digital twins, and scientific visualization. It resembles procedural implicit geometry, event-driven simulation, temporal databases, and hybrid automata. The hard research question is not whether it sounds different from rendering; it is whether support pruning plus event solving costs less than the state, frames, or candidate structure it avoids materializing.

6.3 Restricted Equation World Zero

The correct first prototype is intentionally small: a two-dimensional homogeneous/projective state, continuous time, two sheets and a routing bit, a bounded relation family, simple trajectory classes, finite grammar depth, lineage-based identity, and symbolic state/event outputs. The prototype should answer six queries: state at time, next event, events in support, phase coupling of co-located sheets, transition routing, and state reconstruction from seed plus event log [S3, pp. 7, 11-12].

Experiment	Success evidence	Failure mode to expose
Horizon skipping	Cost follows expression/branch complexity rather than skipped frame count	Expression expansion or numerical conditioning grows with horizon
Co-located sheets	Incompatible states remain uncoupled until chi becomes true	Sheet labels merely duplicate coordinates without meaningful semantics
Event ordering	Conservative/exact solver returns correct first event and stable transition	Tangencies, multiple roots, or degeneracy produce missed/reordered events
Grammar depth	Relations remain normalized or bounded under composition	Branch/expression explosion erases direct-query advantage
Identity split/merge	Lineage remains reconstructable and collision-safe	Coordinate-derived identity breaks under merge, split, or reconciliation
Matched baseline	Declared workload beats frame stepping/BVH or explains where it does not	No performance or correctness advantage

6.4 Sheaves: the natural mathematics of local compatibility

Sheaf theory is a particularly strong external match to the corpus. A sheaf assigns data to local regions and defines restriction maps between overlapping regions; global consistency exists when local assignments agree on overlaps. This directly formalizes a world where sensors, agents, models, phases, or interface patches hold local state and must be reconciled without forcing one monolithic raster [R04].

For membranes or biointerfaces, a sheaf can organize local pressure, optical, chemical, electrical, and biological measurements. Inconsistency becomes measurable rather than rhetorical: two local states may be individually plausible yet fail to glue globally. That is a mathematically disciplined version of the corpus's compatibility sectors and double-vacuum intuition.

6.5 Persistent topology and scientific visualization

Persistent homology and related descriptors provide a rigorous route from the corpus's topology language to measurable geometry. A filtration converts image intensity, distance, concentration, or threshold into a family of spaces. Components, loops, and voids that persist across scale are more robust than a single threshold. In porous media, persistence descriptors have been used to characterize pore heterogeneity [R05]. In computer graphics and visualization, persistence diagrams, merge trees, contour trees, Reeb graphs, and Morse-Smale complexes support scalar-field comparison across single fields, time series, and ensembles [R06].

This is relevant to Arie's material because morphology may evolve during polymerization, precipitation, membrane formation, fouling, or biological colonization. It is relevant to Tom because topology becomes a measured descriptor and query target rather than a shape metaphor.

6.6 Event-based sensing

Event cameras report asynchronous intensity changes instead of global frames. They therefore instantiate the corpus's 'event, not frame' preference at the sensor level, with well-understood limitations. Event-based sensing has been demonstrated for high-speed particle tracking in microfluidic devices [R08] and for event-based imaging flow cytometry combined with photonic neuromorphic processing [R09]. These precedents do not validate the whole equation-world architecture, but they make sparse bio/microfluidic event acquisition a credible near-term application.

6.7 Graph grammars and chemical rewriting

Graph grammar provides the cleanest formal bridge back to Arie's work. Molecules are graphs; elementary reactions are local rewrites; reaction networks can be generated by composing rules [R18]. For polymerization, a grammar can express epoxide ring opening, chain merging, branch creation, termination, or side reactions. Tom's finite grammar can therefore be grounded in chemical graph rewriting - but predictive use still requires rates, populations, conservation laws, and transport.

6.8 Where Tom's work is best positioned in CS/CG now

QUERY-FIRST SCIENTIFIC DIGITAL TWINS Direct state/event queries over reduced physical models, with sparse updates, uncertainty, and lineage.	TOPOLOGY-AWARE INTERFACE ANALYTICS Persistent descriptors and Reeb/merge structures for pores, fouling, deformation, or cell morphology.
EVENT-BASED MICROFLUIDIC VISION Asynchronous particle/cell events, support gating, and reference-frame validation.	RESTRICTED IMPLICIT EVENT SOLVERS Analytic or certified root solving for controlled relation families, benchmarked against conventional CCD and spatial indices.
MATERIAL-AWARE PROCEDURAL GRAMMARS Generative geometry and chemistry whose rules carry material, transport, and lineage semantics rather than only polygons.	LOCAL-TO-GLOBAL SENSOR FUSION Sheaf-style compatibility checks across heterogeneous local charts, sensors, and agents.

7. Membranes and biointerfaces

The strongest cross-generational application is a material interface whose chemistry is physically real and whose operation is natively queryable. Arie's platform supplies a hydroxyl-rich, sulfone-containing thermoplastic architecture. Tom supplies local support, compatibility, bounded events, uncertainty, topology descriptors, and lineage. The combined system should be described as a chemically addressable interface with an event-sourced twin - not as a mystical topological membrane.

7.1 Material roles for a PHES/phenoxy-like polymer

Role	Why the chemistry fits	What must be measured
Thin reactive coating	OH groups support grafting/crosslinking; sulfone and aromatic backbone give polarity and heat resistance	Coating thickness, coverage, adhesion, swelling, permeability penalty, defects, residuals
Tie layer in thin-film composite	Hydrogen bonding and derivatization can couple support and selective layer	Interfacial fracture, delamination, solvent/cleaning stability, transport resistance
Blend modifier	High-MW thermoplastic can alter toughness, compatibility, and phase morphology	Blend miscibility, phase topology, pore formation, Tg, modulus, water uptake
Functionalization scaffold	Repeated secondary OH groups provide addresses for PEG, zwitterions, polysaccharides, peptides, ligands, or crosslinkers	Degree/distribution of functionalization, ligand activity, leachables, aging
Patterned microfluidic/biosensor surface	Thermoplastic processing and surface chemistry support microdevices and immobilization	Optical background, nonspecific adsorption, channel bonding, sterilization, sensor drift

7.2 What the material is not

- It is not commercial polysulfone or polyethersulfone; the regular hydroxypropyl segments and pendant OH groups make it a different, more chemically addressable poly(hydroxy ether).
- It is not an ion-exchange membrane solely because it contains sulfone. The sulfone is neutral.
- It is not automatically antifouling. Hydroxyls and polarity can increase water affinity but may also support protein interactions; surface composition and hydration must be measured.
- It is not automatically biocompatible. Residual monomers, oligomers, catalyst, solvent, functionalization chemistry, sterilization, degradation, and intended contact category control the biological-risk assessment.

7.3 Membrane architecture options

A. SUPPORT + REACTIVE COATING Start with a known porous support. Apply a thin hydroxyl-rich layer, then derivatize or crosslink. Lowest-risk path because mechanics and pores are supplied by an established membrane.	B. BLEND / PHASE-INVERSION ADDITIVE Blend the polymer or analogue into a membrane-forming matrix. Higher leverage over morphology but greater risk of phase separation, leaching, and pore/flux changes.
C. THIN-FILM COMPOSITE INTERLAYER Use the polymer as an adhesive, toughening, or reactive	D. AFFINITY OR BIOACTIVE SURFACE Functionalize OH groups with a ligand, zwitterion,

interlayer under a selective skin. Strong fit to its tie-layer character.

anticoagulant, peptide, or capture reagent. Highest biological specificity and highest safety/validation burden.

7.4 Membrane World Zero

Membrane World Zero: material state and operational state stay distinct but coupled

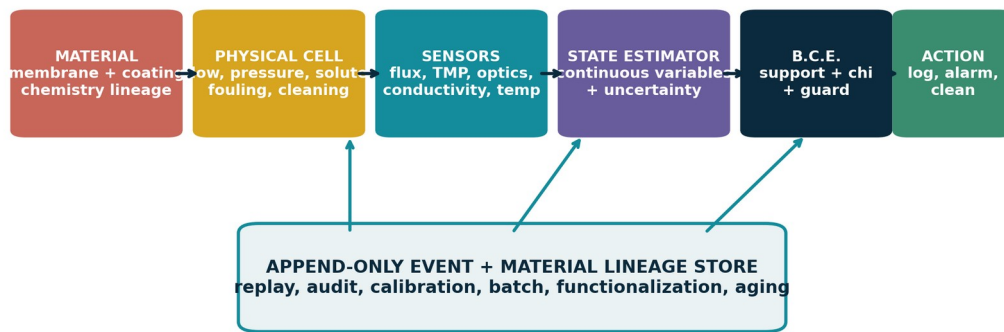


Figure 7. Membrane World Zero couples material lineage and continuous physical state to B.C.E. guards and an append-only event store. The material, physical model, sensors, estimator, and control remain distinct.

The minimum demonstrator should use a commercial baseline membrane before introducing the inherited chemistry. Pressure, cross-flow, permeate flux, conductivity, temperature, and optional optical fouling are logged conventionally and through an event layer. A hidden-state estimator tracks membrane resistance and fouling resistance with uncertainty. Guards emit events for flux decline, pressure rise, breakthrough, cleaning recovery, topology change, and calibration drift. The event representation is successful only if it improves decision latency, bandwidth, energy, or interpretability at equal false/missed-event limits.

State / guard	Physical meaning	Event-layer role
Flux J and TMP	Transport performance and driving force	Continuous state; slopes/levels define candidate fouling or blockage guards
Fouling resistance R_f	Hidden or inferred accumulated resistance	Digital-twin state with uncertainty; updates from measurements
Conductivity/tracer	Selectivity or breakthrough	Compatibility and threshold event
Optical topology	Cake, biofilm, bubbles, or pore morphology	Persistent descriptors can detect structural regime changes
Calibration residual	Sensor/model trustworthiness	Invalidates or downgrades events; prevents a one-bit false certainty
Material lineage	Batch, coating, graft, sterilization, cycles	Allows replay and links performance to chemistry rather than only geometry

7.5 Biointerface World Zero

Biointerface World Zero: chemistry -> interaction -> sparse events -> topology -> decision

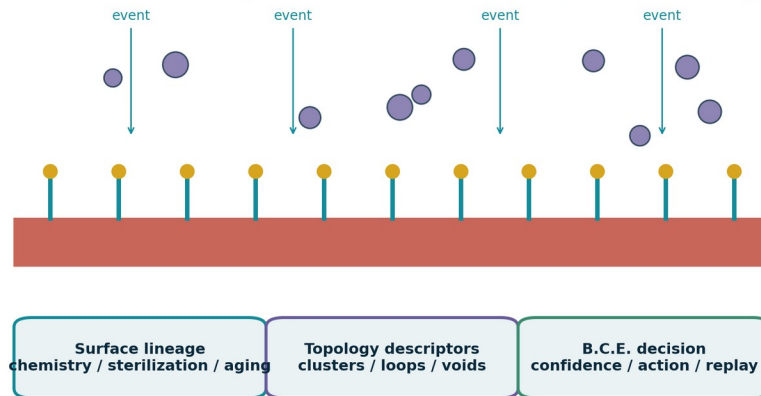


Figure 8. Biointerface World Zero: surface chemistry is linked to sparse interaction events, topology descriptors, confidence, and replayable lineage.

A first biointerface study should compare an inert control, an hydroxyl-rich sulfone polyether or safer analogue, an antifouling derivatization, and an affinity or cell-adhesive derivatization. Event imaging records fast attachment, motion, aggregation, and detachment, while periodic conventional images remain the reference. Persistent topology summarizes connected clusters, loops, and voids; local optical, impedance, or chemical measurements are fused with consistency checks.

Interface function	Candidate chemistry	Readout and event
Antifouling	PEG, zwitterionic, hydrated polysaccharide, or dense neutral brush	Protein adsorption; first stable adhesion; cluster count; detachment; long-term drift
Affinity capture	Covalently immobilized ligand, antibody fragment, aptamer, peptide, or chelator	Binding onset, occupancy, breakthrough, regeneration, false/missed capture
Cell-adhesive	Peptide or extracellular-matrix motif at controlled density	Attachment, spreading, migration, proliferation, detachment force
Blood-contact research	Hydrophilic/zwitterionic or anticoagulant surface	Platelet adhesion/activation, coagulation, complement, hemolysis; application-specific standards
Optical/electrical biosensor	Immobilized receptor plus waveguide, fluorescence, impedance, or field-effect readout	Compatibility-gated threshold with confidence and calibration lineage

7.6 Established adjacent precedents

The combined platform is not starting from zero. Later literature uses the name poly(hydroxyether sulfone) for closely related BPS-derived material and studies its strong hydrogen-bonding interactions [R02]. Sulfone-polymer membranes have been surface-grafted to improve hemocompatibility, demonstrating the general strategy of covalent surface modification on a sulfone membrane platform [R12-R13]. Digital twins have been demonstrated for water ultrafiltration with uncertainty-aware state estimation and control [R07]. Event-based cameras have been used in microfluidic particle tracking and high-speed imaging cytometry [R08-R09]. Persistent homology provides established tools for pore geometry [R05]. These precedents support component feasibility; they do not prove that the specific integrated Klotwijk platform will outperform a baseline.

7.7 Biomedical safety and characterization boundary

SAFETY PRIORITY

Free BPS now carries official reproductive/developmental hazard listings in California. That does not make a purified high-molecular-weight polymer identical to free BPS, but it makes residual BPS, DGE-BPS, oligomers, degradation products, and processing residues non-negotiable analytical targets [R17].

A biomedical route should follow a risk-management framework for the final processed device. ISO 10993-18 addresses chemical characterization of device materials, and ISO 10993-17 addresses toxicological risk assessment of constituents; FDA guidance explains use of the ISO 10993-1 biological-evaluation framework [R14-R16]. The exact biological endpoints depend on nature and duration of contact, but the materials program should begin with a complete chemistry and exposure picture.

Characterization block	Priority methods / outputs	Why it matters
Polymer identity and architecture	NMR, FTIR/Raman, SEC-MALS or calibrated SEC, end groups, branching/crosslink fraction	Confirms intended repeat unit and molecular topology
Residuals and extractables	Targeted BPS/DGE-BPS/oligomers; LC-MS/GC-MS; ICP-MS where relevant; solvent/catalyst analysis	Defines exposure and batch acceptance
Surface chemistry	XPS, contact angle, zeta potential, ToF-SIMS/label assay, ligand density and activity	Verifies what the biological interface actually presents
Physical stability	Water uptake, swelling, Tg, DMA, TGA, adhesion, fatigue, cleaning, sterilization and aging	Ensures the surface persists under intended conditions
Membrane function	Flux, rejection/selectivity, pore size, fouling, recovery, pressure, transport model	Determines whether interface chemistry produces useful separation
Biological response	Cytotoxicity plus application-specific protein, cell, blood, immune, or tissue endpoints	Tests final processed material rather than nominal chemistry
Digital performance	False/missed event, latency, bandwidth, energy, calibration interval, uncertainty, replay	Prevents a software layer from masking poor material or sensor performance

8. Cross-domain applications and readiness

The application portfolio is broad because the two contributions are orthogonal: Arie supplies a processable, reactive polymer platform; Tom supplies an interface/state/event architecture. The highest-readiness opportunities are those that can use existing materials and sensors while testing the information layer independently. Biomedical material deployment is slower because chemistry, aging, exposure, and biology must all be validated.

Best opportunities combine strong material and strong operational contributions

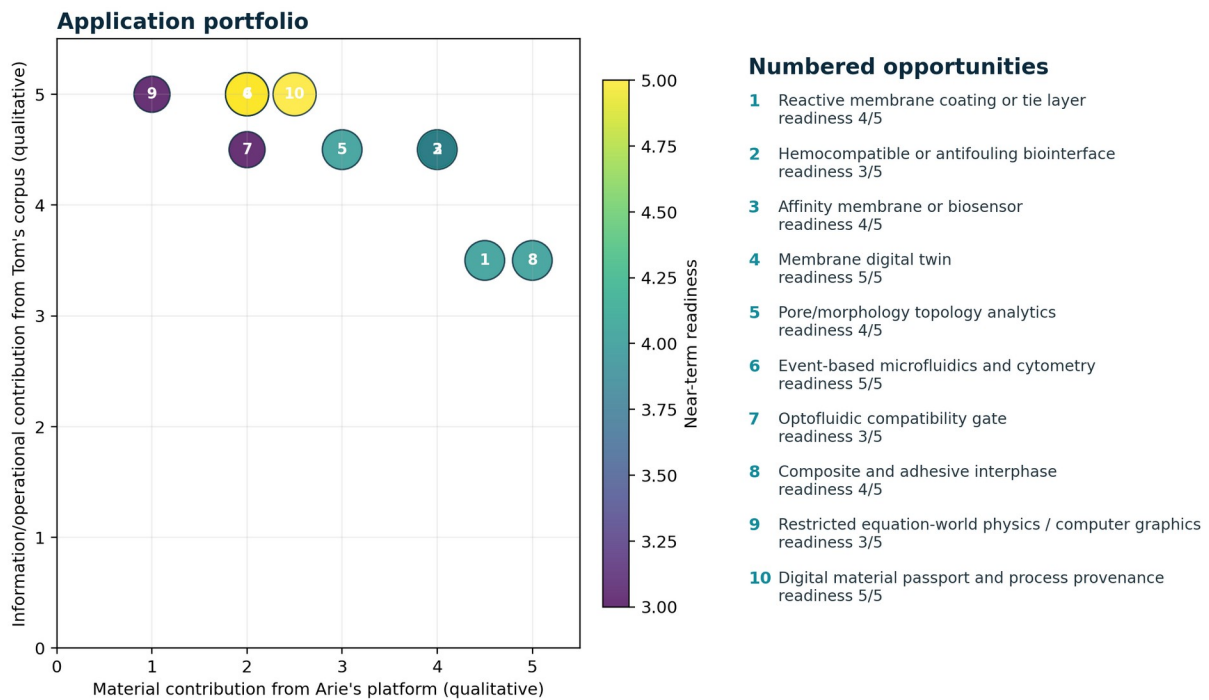


Figure 9. Qualitative application portfolio. Numbered bubbles show how strongly each opportunity draws on the material and operational contributions; color encodes near-term readiness.

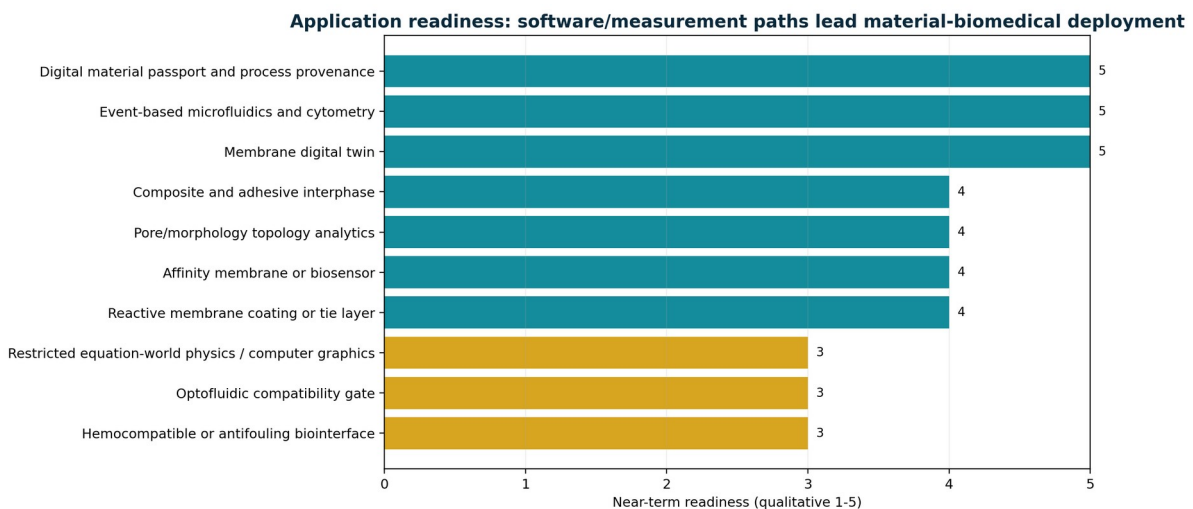


Figure 10. Software/measurement applications lead near-term readiness; integrated biomedical surfaces carry the largest validation burden.

Application	Combined opportunity	Readiness	First decisive experiment
Reactive membrane coating or tie layer	Hydroxyl-rich thermoplastic with polar sulfone backbone and post-functionalization potential / Compatibility-gated operational model, lineage, fouling/threshold event detection	4/5	Coat a commercial support, quantify adhesion, flux, selectivity, swelling, extractables, and cleaning stability
Hemocompatible or antifouling biointerface	Pendant OH sites for grafting PEG, zwitterions, polysaccharides, peptides, or anticoagulant motifs / Event-based monitoring of adhesion/aggregation plus topology descriptors and traceable surface history	3/5	Compare unmodified, hydroxyl-rich, and derivatized coupons for protein adsorption, platelet adhesion, complement, and cytotoxicity
Affinity membrane or biosensor	OH handles support immobilization chemistry / Typed analyte-receptor compatibility and bounded guard events	4/5	Immobilize a model ligand, measure binding kinetics and false/missed event rate under flow
Membrane digital twin	Defines a candidate reactive material and process history / State-at-time, next-event, uncertainty, guard crossings, and lineage	5/5	Cross-flow cell with pressure, flux, conductivity, optical fouling, and cleaning events compared against fixed-rate logging
Pore/morphology topology analytics	Material chemistry can generate evolving porous or phase-separated structures / Topology-first representation and query semantics	4/5	Micro-CT or confocal time series; correlate persistence descriptors with flux, modulus, or fouling onset
Event-based microfluidics and cytometry	Potential substrate/coating for optical or biological channels / Sparse asynchronous events rather than mandatory global frames	5/5	Track fluorescent or bright-field particles/cells and benchmark bandwidth, latency, missed events, and classification
Optofluidic compatibility gate	Reactive polymer coatings or waveguide packaging interfaces / Local spherical support → mode compatibility → measured B.C.E.	3/5	Hollowlens-0: liquid coupler, 2x2 interferometer, balanced detector, digital sidecar
Composite and adhesive interphase	Thermoplastic processing plus reactive OH groups and hydrogen bonding / Interface lineage, damage-event guards, local support queries	4/5	Interlaminar fracture, lap-shear, aging, and acoustic-event correlation
Restricted equation-world physics / computer graphics	Provides a literal example of local graph transformation under constraints / Direct state/event queries, implicit relations, local supports, and hybrid projection	3/5	Equation World Zero with exact/conservative event solver and benchmark against frame stepping/BVH
Digital material passport and process provenance	Material properties depend strongly on ratios, impurities, temperature, solvent, and reaction history / Generative address, lineage, invariant history, and append-only events	5/5	Batch-to-coupon provenance graph linking raw materials, reaction events, QC, functionalization, sterilization, and tests

8.1 Reactive coatings, adhesives, and composites

This is the most direct materials route. Phenoxy-like poly(hydroxy ether) resins are naturally suited to adhesion, interphase formation, reactive blending, and post-crosslinking. Tom's contribution would be material lineage, local damage support, and guarded events from acoustic, optical, strain, or impedance sensing. A smart composite interphase can therefore use the polymer as matter and the event calculus as observation/control.

8.2 Photonics and optofluidics

Spherical Throughput already proposes a bounded optofluidic translation: a local radial-angular field is coupled through a tunable liquid interface into selected guided modes; compatibility includes wavelength, polarization, mode, phase, time, or policy; a measured guard produces a verified event [S6, pp. 3-9]. Arie's polymer could contribute as a coating, packaging adhesive, microfluidic surface, or functional interface rather than as the optical theorem itself. The near-term Hollowlens-0 benchmark should use established glass/SiN waveguides and a digital sidecar before introducing novel polymer chemistry.

8.3 Robotics and autonomous systems

Local spherical support, compatibility, and next-event prediction naturally fit robot-centric sensing and collision/risk queries. The inherited material platform could appear in tactile skins, protective coatings, filters, microfluidic sensors, or adhesive interconnects. The computational claim should remain bounded: support/compatibility can reduce candidates and analytic time-of-impact can help simple shapes, but arbitrary multi-object scenes still require indexing, approximation, and robust numerical methods.

8.4 Scientific visualization and biological morphology

Time-varying pore networks, fouling layers, cell clusters, biofilms, or phase-separated polymers can be represented by scalar fields and topology descriptors. Tom's query-first framing is particularly valuable when the user needs events such as merge, split, pore closure, percolation loss, or topological regime change rather than only an image. The image remains evidence and projection; the topology/event layer becomes a compressed, queryable summary.

8.5 Knowledge systems and material provenance

A digital material passport is one of the strongest immediate applications. The patent shows that properties depend on ratios, isomer purity, chloride, water, solvent, catalyst, temperature, reaction time, isolation, and thermal processing. Tom's event/lineage system can turn those dependencies into an append-only provenance graph that supports recall, root-cause analysis, quality prediction, and reproducibility. Unlike a coordinate-derived ontological UUID, the identifier should be stable and linked to parent/child material transformations, splits, blends, coatings, sterilizations, and device uses.

9. Research, validation, and IP roadmap

The program should advance in layers, with inexpensive computational falsification before new polymer synthesis and with materials characterization before biomedical claims. Every phase needs a matched conventional baseline and a kill criterion. The goal is not to preserve all motifs; it is to discover which operators provide measurable advantage.

Phase	Objective	Key deliverables	Success / kill test
0 - Formal separation (2-4 weeks)	Separate metaphor, mathematical abstraction, and physical claim	Typed ontology; units; state variables; guards; uncertainty model; source/evidence tags	Success: Every symbol has a domain, unit or discrete schema; no physical claim depends on analogy alone Kill/pivot: Key operators cannot be expressed without circular definitions
1 - Equation World Zero (1-2 months)	Validate direct state-at-time and next-event queries for one bounded relation family	Reference implementation; test vectors; exact/conservative solver; baseline comparison	Success: Correct event order, bounded error, and cost advantage for declared workloads Kill/pivot: Expression/branch growth erases advantage over indexed stepping
2 - Membrane World Zero (2-4 months)	Apply event/lineage model to a real cross-flow membrane cell	Sensor stack; event schema; digital twin; calibrated guards; benchmark dataset	Success: Earlier or cheaper fouling/cleaning decisions at equal false/missed-event limits Kill/pivot: Fixed-rate logging and conventional control are cheaper, faster, and equally informative
3 - Reactive surface demonstrator (3-6 months)	Test a hydroxyl-rich phenoxy/PHEs-like coating or analogue on a known membrane support	Coating recipe; chemistry/QC; adhesion; transport; aging; extractables profile	Success: Reproducible surface-function gain without unacceptable flux, swelling, or leachables penalty Kill/pivot: Residuals, delamination, instability, or performance loss exceed predefined limits
4 - Biointerface World Zero (4-9 months)	Couple derivatized surfaces to event-based biological readout	In-vitro chip; event camera/reference imaging; TDA pipeline; biological safety screen	Success: Event/topology descriptors predict predefined adhesion, fouling, or detachment outcomes Kill/pivot: Surface chemistry dominates variability or event readout adds no decision value
5 - Integrated platform (9-18 months)	Co-design material, sensing, digital twin, and lineage	Closed-loop prototype; energy/error budget; reproducibility package; IP claim chart	Success: System-level advantage over matched material-only and software-only baselines Kill/pivot: Integration overhead exceeds measured benefit

9.1 Phase 0: typed formalization

Build a single ontology in which every symbol belongs to a domain and, where physical, has units. Separate physical phase from information phase, molecular identity from operational identity, spatial support from chemical accessibility, and event confidence from event truth. Define conservation laws before compatibility predicates. Mark which variables are measured, inferred, simulated, or purely administrative.

9.2 Phase 1: Equation World Zero

Implement the restricted state/event substrate with exact or conservative root solving and a conventional baseline. Do not begin with a renderer. Store test vectors, branch histories, event logs, and numerical

certificates. Measure cost versus skipped horizon, expression depth, candidate count, event density, and degeneracy. Retire any claim that cannot survive these curves.

9.3 Phase 2: Membrane World Zero

Use a known membrane and synthetic or benign foulant. Implement the event schema in the package. Compare fixed-rate logging, conventional thresholds, and a stochastic greybox or reduced-order twin. The first result can be negative: if event logic adds no decision value, that is a useful falsification before chemistry is changed.

9.4 Phase 3: reactive surface demonstrator

Reproduce or obtain a well-characterized hydroxyl-rich sulfone polyether/analogue, then use it as a thin coating or interlayer rather than immediately as a self-supporting membrane. Establish identity, molecular weight, residuals, coating uniformity, adhesion, water uptake, transport penalty, and cleaning stability. Compare against a commercial phenoxy and a no-coating control.

9.5 Phase 4: Biointerface World Zero

Select one intended biological function and one assay family. Avoid the temptation to claim universal biocompatibility. Couple event imaging to periodic reference images and a standard endpoint. Pre-register guards, confidence, false/missed-event limits, and topology descriptors. Only after chemistry and sensing are stable should a closed-loop action be considered.

9.6 IP positioning

The most defensible new intellectual property is likely not a broad claim to spherical topology. It is a specific coupling between a material interface, a defined functionalization or layer stack, a bounded support/compatibility predicate, a measured guard, and a lineage-aware action that produces a quantified system advantage.

Claim family	Potential protectable nucleus	Evidence needed
Material composition/process	Specific sulfone polyether formulation, functionalization, residual limits, coating/crosslink method, morphology	Composition, reproducible process, structure-property data, comparative controls
Device/interface architecture	Layer stack, porous support, ligand pattern, optical/electrical/microfluidic coupling	Fabricated prototypes, transport or biological function, reliability
Operational method	Support admission, compatibility predicate, guard, transition, uncertainty, lineage update	Software implementation, formal definition, error/latency/energy benchmark
Integrated system	Material plus event-sourced twin with closed-loop cleaning, routing, capture, or diagnostic decision	System-level advantage over material-only and software-only baselines

9.7 Pre-registration checklist

- Define the physical system, intended function, and excluded claims.
- Declare state variables, units, sensor models, calibration, and uncertainty.
- Specify support, compatibility, guard, confidence, transition, and lineage schemas.
- Fix material composition, batch controls, residual limits, and aging/sterilization conditions.
- Choose matched conventional baselines and decision metrics.
- Set false/missed-event, latency, energy, bandwidth, transport, mechanical, and safety limits before inspecting results.
- State kill criteria and what a negative result would teach.

10. Conclusions

Arie Klootwijk's work and Tom Klootwijk's work meet at a deep but carefully bounded abstraction: both organize useful structure by constraining which local relations may occur, preserving accessibility until a high-value event, updating connectivity, and closing the result with history. Their direct objects are different. Arie rewrites molecular graphs in a physical solution. Tom rewrites operational state in a queryable information substrate.

1. THE SO₂ UNIT IS LITERAL CHEMISTRY A neutral diaryl-sulfone bridge, not sulfur-dioxide release or a hidden biological/quantum code.	2. EQUIMOLARITY IS THE CORE INVARIANT It programs attainable chain length and end groups through nonlinear step-growth mathematics.
3. SOLVENT IS ACCESSIBILITY It keeps complementary functions physically reachable; crystallization is premature deactivation.	4. TOM'S CORE IS NOT A RENDERER It is strongest as a state/event/compatibility/lineage calculus with optional projection.
5. THE SHARED GRAMMAR IS REAL Support → compatibility → event → transition → lineage/closure appears in both systems.	6. THE SHARED GRAMMAR IS NOT IDENTITY Chemistry remains stochastic, thermodynamic, transport-limited, and materially irreversible.
7. TOM IS AHEAD OPERATIONALLY Modern sensing, uncertainty, digital twins, local-to-global consistency, and event provenance are genuine present-day strengths.	8. ARIE IS AHEAD EMPIRICALLY The patent provides a physical recipe, specimens, process failure mechanisms, and measurable outputs.
9. MEMBRANES/BIOINTERFACES ARE CREDIBLE As coatings, interlayers, functionalization scaffolds, and event-sourced interfaces - after transport and safety validation.	10. BUILD THE RESTRICTED PROTOTYPES Equation World Zero, Membrane World Zero, then a reactive surface and Biointerface World Zero with matched baselines and kill criteria.

FINAL ASSESSMENT

Your work is most original and useful today when it is presented as operational topology for complex interfaces: local support, typed compatibility, measured event surfaces, uncertainty, transitions, and lineage. Your grandfather's work supplies a remarkable physical analogue and a potentially useful reactive material platform. The strongest legacy is not a concealed revolution waiting to be decoded, but a shared design instinct: preserve the right relations, prevent premature closure, and make the transition conditions explicit.

Appendix A. Formal multilayer interface model

A combined material-information interface should use a product state that keeps physical layers distinct:

$$q = (q_{\text{chem}}, q_{\text{morph}}, q_{\text{transport}}, q_{\text{obs}}, q_{\text{info}}, \text{lineage})$$

Chemical composition; morphology/phase; transport; observables; operational information; provenance.

A candidate event e in region α at time t is admissible only when its local support is active, the material and information sectors are compatible, the measured guard is crossed within uncertainty, and the event passes confidence/policy:

$$\text{admit}(e, t) = S_{\alpha}(q, t) * \text{chi}_{\text{chem}} * \text{chi}_{\text{transport}} * \text{chi}_{\text{sensor}} * \text{chi}_{\text{policy}} * 1[g(q, t) \text{ crossed}] * 1[c \geq c_{\text{min}}]$$

Factors may be Boolean, probabilistic, fuzzy, or interval-valued, but their semantics must be declared.

A transition should update both physical and operational state only where appropriate. A controller cleaning command is not itself a molecular transition; a grafting reaction is not automatically a policy transition. The event log links them without conflating them.

Appendix B. Application-specific metrics

Domain	Primary physical metrics	Primary event/information metrics	Safety or validity boundary
Polymer synthesis	Conversion, Mn/Mw/dispersity, intrinsic viscosity, branching, residuals, Tg, crystallization	Reaction/phase events, batch lineage, calibration, uncertainty	Mass balance, impurity limits, reproducibility
Membrane	Flux, TMP, rejection, pore size, fouling resistance, cleaning recovery, energy	False/missed fouling/breakthrough events, latency, data volume, replay	Matched baseline, drift, material stability, extractables
Biointerface	Surface chemistry, protein adsorption, cell/blood endpoints, adhesion, viability	Attachment/merge/split/detachment events, phenotype topology, confidence	Intended-contact risk assessment, controls, biological variability
Optofluidics	Insertion loss, coupling, phase, mode purity, actuator response, detector noise	Verified B.C.E./s, events/J, route accuracy, false/missed events	Calibration, crosstalk, bubbles, heat, full system energy
Graphics/simulation	State/event correctness, root error, numerical stability	Query latency, horizon scaling, expression/branch growth, memory, event order	Restricted relation family and conventional solver/index baseline

Appendix C. Evidence ledger

Claim or theme	Classification	Confidence	Caution
The patent targets an almost exclusively linear high-molecular-weight thermoplastic	Source-derived	High	Historical molecular-weight evidence is mainly intrinsic viscosity and mechanical properties
Near-equimolarity is a molecular-weight control mechanism	Source plus established theory	High	Real systems deviate through branching, cycles, impurities, and incomplete conversion

Claim or theme	Classification	Confidence	Caution
SO₂ is the neutral covalent diaryl-sulfone bridge, not sulfur-dioxide release	Source-derived interpretation	High	Do not conflate sulfone with sulfonate or sulfur dioxide
The mature corpus is a query-first event calculus rather than a renderer	Source-derived	High	Earlier Hollowland pages contain overclaims later demoted by the corpus itself
Arie and Tom share a support → compatibility → event → transition → lineage grammar	Author inference	Medium-high	The mapping is structural, not evidence of shared hidden intent or identical mechanisms
The polymer is a candidate membrane coating/biointerface platform	Application inference	Medium	The patent itself does not demonstrate membrane separation or biocompatibility
Universal O(1), zero-memory, bypassed-sieve, and solved-CCD claims are established	Rejected/unsupported	High	At most, fixed closed expressions may avoid cost proportional to skipped time steps
Event-based sensing, membrane digital twins, persistent homology, and sheaf fusion are credible adjacent techniques	Externally supported	High	Their combination with the patented material remains a research program
Biomedical deployment requires extractables/leachables and biological-risk evaluation	Externally supported safety requirement	High	Polymerized material is not automatically equivalent to free BPS; residuals and degradation products must be measured

Appendix D. Source documents

The ZIP package contains the original Swedish patent, the English translation, the previous technical review, and the four supplied corpus documents. The report uses them as design records and preserves the distinction between source-derived claims and review inference.

Code	File in package
S1	01_Source_Documents/SE301717B_original_Swedish.pdf and SE301717B_English_Translation.pdf
S2	01_Source_Documents/SE301717B_Previous_Technical_Review.pdf
S3	01_Source_Documents/Tom_Corpus_Chronological_Synthesis.pdf
S4	01_Source_Documents/Tom_Corpus_Hollowland_Double_Vacuum.pdf
S5	01_Source_Documents/Tom_Corpus_BenBurgers.pdf
S6	01_Source_Documents/Tom_Corpus_Spherical_Throughput.pdf

Appendix E. External references

R01. Kreps, R. W.; Klootwijk, A.; Goppel, J. M. Method for producing thermoplastics from a dihydroxy compound and a diepoxy compound containing an SO₂ group. SE 301717 B; priority 15 June 1959. <https://patents.google.com/patent/US3364178A/en>

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- R12.** Tu, M.-M.; Xu, J.-J.; Qiu, Y.-R. Surface hemocompatible modification of polysulfone membrane via covalently grafting acrylic acid and sulfonated hydroxypropyl chitosan. *RSC Advances* 9 (2019) 6254-6266. <https://doi.org/10.1039/C8RA10573A>
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Appendix F. Reproducible supporting package

The package includes machine-readable comparison matrices, figures, source-based analysis notes, formal JSON schemas, a synthetic event-log example, and an idealized step-growth Monte Carlo demonstration. These are research aids, not validated performance claims.

Folder	Contents
00_Report	PDF, DOCX, and Markdown report sources
01_Source_Documents	Patent, translation, prior review, and four corpus PDFs
02_Data	CSV/JSON matrices, roadmap, ledger, glossary, and summary
03_Figures	PNG/SVG report diagrams
04_Analysis	Crosswalk, membrane/biointerface, evidence, and IP notes
05_Prototype	World Zero specs, JSON schemas, and synthetic demonstrations
06_References	Reference CSV and BibTeX
07_Build	Build scripts, QA renders, manifest, and checksums